

Topic: Recursive Queries

Q. What are Recursive Queries in SQL Explain the concept of Common Table Expressions (CTE) and how a Recursive CTE works with an example.

[DEF] Definition to Remember

A Recursive Query is a query that repeatedly executes itself until a termination condition is met, used to traverse hierarchical or tree-structured data. It is implemented in SQL using a Recursive CTE (Common Table Expression) with the WITH RECURSIVE keyword, consisting of an Anchor Member and a Recursive Member joined by UNION ALL.

Recursive Queries & Common Table Expressions (CTE)

A CTE is a temporary, named result set defined using the WITH keyword. It exists only during the execution of a single SQL statement.

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2. Structure of a Recursive CTE

A recursive CTE has exactly 3 components:

```
WITH RECURSIVE CTE_Name AS (

    ANCHOR MEMBER                <- Base case (executes once, gives R )
    SELECT ...

    UNION ALL                    <- Connects anchor + recursive

    RECURSIVE MEMBER            <- References CTE itself (gives R , R , ...)
    SELECT ... JOIN CTE_Name ...

)
SELECT * FROM CTE_Name;
```

3. Working of a Recursive Query

1. Anchor Member executes first -> produces base result set R₀.

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4. Example -- Organizational Hierarchy

Table: Employee (EmpID, Name, ManagerID) -- Find all levels of management.

```
WITH RECURSIVE
  --
  --
  SE
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  WH
  UN
  --
  SE
  FR
  IN
  )
SELECT
```

5. Applications

* Organizational Charts: Employee hierarchy traversal.

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[MUST-WRITE] Must-Write Points (for 10 marks)

1. Recursive Queries are used for hierarchical or tree-structured data (org charts, BoM).
2. Implemented using Recursive CTE with the WITH RECURSIVE keyword.
3. Three components: Anchor Member (base case) + UNION ALL + Recursive Member (self-referencing).
4. Anchor Member executes once; Recursive Member repeats until it returns an empty set.
5. Final result = UNION ALL of all intermediate result sets.
6. CTE is a temporary named result set -- exists only for the duration of one query.
7. Without Recursive CTEs, hierarchical queries required complex procedural code loops.

[RECALL] Quick Recall

Recursive Query -> Hierarchical Data -> WITH RECURSIVE -> Anchor (base R) -> UNION ALL -> Recursive Member (R , R ...) -> Terminates when empty -> Apps: Org Charts, BoM, Routing